GAURAV ARORA

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PROFESSIONAL SUMMARY

Senior QA Lead / Automation Engineer (SDET) with **13+ years of experience** in building scalable automation frameworks and driving end-to-end quality engineering across cloud-native and enterprise applications.

Strong expertise in **Test Automation, Selenium, Java, Playwright, TypeScript , API testing**, combined with hands-on experience in **Linux, Kubernetes, Terraform, CI/CD pipelines, OCI and AWS** cloud platforms.

Recently designed and implemented an **AI-enhanced automation framework** integrating **AI-driven test generation**, **intelligent test data creation, smart locator strategies,** and **self-healing** mechanisms using **fallback** and **mocking approaches** to improve test reliability and reduce maintenance overhead.

Experienced in building resilient automation systems with **AI-based failure analysis, intelligent reporting**, and CI/CD-integrated insights to enable faster and reliable releases in cloud environments (OCI, AWS).

CORE SKILLS

QA Leadership & Strategy: Test Planning, Test Strategy, Test Automation, Stakeholder Management, Agile/Scrum, Cross-team Coordination.

Automation & Testing: Selenium, Java, Playwright, TypeScript, TestNG, JUnit, Maven, REST Assured, Cucumber (BDD), UI Testing, API Testing, Microservices Testing, Automation Framework Design.

Cloud & DevOps: OCI, AWS, Kubernetes, Terraform, Docker, Helm, CI/CD (Jenkins, GitHub Actions, GitLab CI), Continuous Testing.

Systems & Scripting: Linux (Advanced), Linux Administration, Bash Scripting, Debugging, Log Analysis.

Tools & Technologies: Git, Postman, Grafana, DynamoDB, MongoDB, Postgres.

AI-Enhanced Automation: AI-driven Test Data & Scenario Generation; Self-healing Test Design (Fallback & Mocking); Smart Locator Strategies & Dynamic Element Handling; AI-based Failure Analysis & Intelligent Reporting; Data-driven Testing using JSON + TestNG DataProviders.

Cloud-Native Testing: Kubernetes-based microservices testing and validation; Infrastructure validation using Terraform; CI/CD-integrated automation pipelines; Distributed system debugging using Linux and monitoring tools.

PROJECTS

AI-Enhanced Automation Framework

GitHub: github.com/aroragaurav2810/ai-cloud-native-automation-framework

- Developed a **Selenium** + **TestNG** + **Rest Assured** framework integrated with AI capabilities across the testing lifecycle.
- Implemented **AI-driven test data generation and test scenario generation** to improve test coverage.
- Designed **fallback** and mocking mechanisms to ensure resilience when AI services are unavailable.
- Enabled data-driven execution using **AI-generated JSON datasets** with TestNG DataProviders.
- Integrated AI-based failure insights and intelligent reporting into CI/CD pipelines.
- Containerized framework using Docker and designed for scalable execution.

Terraform Demo Infrastructure Framework

GitHub: github.com/aroragaurav2810/terraform-demo-infra-setup

- **Terraform**-based framework to provision and manage demo lab infrastructure for **Oracle Access Governance Cloud Service (AGCS)**, enabling consistent, on-demand environment setup before customer demos.
- Provisioned infrastructure to demonstrate Oracle Identity & Access Governance connectors - including OID (Oracle Internet Directory), OUD (Oracle Unified Directory), EBS (Oracle E-Business Suite), Microsoft Active Directory, NetSuite, Oracle Fusion Applications, MySQL, and MSSQL - against a live AGCS tenant for end-to-end validation.

PROFESSIONAL EXPERIENCE

Oracle - Principal Member of Technical Staff

QA Lead / Automation & Cloud Testing | Sep 2022 – Present

Project: Oracle Access Governance (AGCS), OCI Security Initiative - SpeedDial

- Led QA initiatives for cloud-native Identity Management products (AGCS) across distributed environments.
- Designed and implemented **AI-enhanced automation frameworks** incorporating **AI-driven test generation**, intelligent test data, and **smart locator strategies** using **Selenium**, **Java**, and **TestNG**.
- Build scalable **UI** automation framework using **Playwright** and **TypeScript**
- Build scalable **API** automation using **Java** and **RestAssured**.

- Led end-to-end connector testing for **OID, OUD, EBS, Microsoft Active Directory, NetSuite, Fusion Applications, MySQL,** and **MSSQL** - covering aggregation, entitlement sync, provisioning, de-provisioning, and access review workflows within AGCS.
- Implemented **self-healing test approaches** using fallback and mocking, ensuring stable execution in CI/CD pipelines despite external service dependencies.
- Integrated **AI-based failure analysis** and **intelligent reporting**, reducing debugging effort and improving defect triage.
- Enabled data-driven testing using **AI-generated datasets** and JSON-based DataProviders.
- Automated infrastructure provisioning using **Terraform**, **reducing environment setup time by 70–80%**.
- Collaborated with Dev, DevOps, and Product teams to streamline CI/CD pipelines and release cycles.
- Performed system-level debugging using **Linux** (log analysis, process monitoring, and troubleshooting).

S&P Global / IHS Markit - SQA Engineer

Senior QA Engineer | May 2017 – Sep 2022

Project: Financial Markets – Loan Solution

- Led automation efforts for financial platforms handling high-volume data systems.
- Built scalable UI and API automation frameworks using Selenium, Java, TestNG, and BDD.
- Implemented mock server-based testing to simulate backend services, reducing dependency on external systems and improving test stability in CI/CD pipelines.
- Developed backend validation frameworks integrating AWS services (Lambda, DynamoDB, ECS).
- Implemented CI/CD pipelines, improving deployment frequency and reducing release failures.

SirionLabs - Senior Test Engineer

Senior Test Engineer | Dec 2014 – May 2017

Project: Sirion (AI Contract Management Platform)

- Improved regression cycles through parallel execution and framework optimization.
- Developed automation frameworks for UI and backend systems using Selenium, Java and RestAssured.

CyberQ Consulting (Client: NIC) - Software Test Engineer

Software Test Engineer | May 2013 – Nov 2014

Project: PRIASoft (Panchayati Raj Institutions Accounting Software)

- Developed UI automation frameworks and conducted API and database testing.

Eastern Horizon Solutions - Software Tester

Software Tester | May 2012 – Apr 2013

Project: gharexpert.com

- Performed manual and regression testing for e-commerce applications.

KEY ACHIEVEMENTS

- Reduced infrastructure setup time by 70–80% using Terraform automation.
- Improved regression efficiency by ~50% through automation frameworks.
- Enabled faster release cycles by integrating QA into CI/CD pipelines.
- Built and scaled Kubernetes-based test environments for microservices.

CERTIFICATIONS

- OCI AI Foundation Associate (2025)
- OCI Foundation Associate (2022, 2023)
- Oracle Java Certified (2017)

EDUCATION

B. Tech (CSE), MDU - 2012